

Rajarata University of Sri Lanka

Department of ICT

ICT 1209 – Web Technologies

Mini Project Guide | 2024 Batch

Mini Project: Interactive Web Application Development

Overview

This mini project contributes 20% of your final assessment for ICT 1209 – Web Technologies. You will design and develop a fully functional, database-driven interactive web application in pairs, progressing through three structured phases across the semester.

Item	Details
Course	ICT 1209 – Web Technologies
Batch	2024 (First Year, Bachelor of ICT)
Group Size	Pairs (2 students per group)
Group Registration Deadline	Friday, 19th June 2026
Total Weightage	20% of final assessment

Required Technology Stack

All groups must use the following technologies. The tech stack is aligned with the approved course syllabus and is designed to assess your competency in these specific fundamentals.

Layer	Technologies
Frontend Structure	HTML5, CSS3, Bootstrap 5
Client-side Logic	JavaScript (Vanilla)
Backend	PHP 8
Database	MySQL (via XAMPP or WAMP)
Version Control	Git & GitHub

Theme Selection

Each group must select have a theme for their web application. **You can propose your own theme** subject to approval. Some **example** themes are listed below:

- Digital Recipe Book – Recipe search, dynamic content, user-submitted recipes
- Fitness Tracker – Activity logging, interactive charts, goal tracking
- Quiz or Trivia Game – Dynamic question generation, score tracking, feedback
- Habit Tracker – Progress visualization, dynamic goal updates

- Student Resource Hub – Upload/browse notes, search, categories
- Any other innovative theme (requires prior approval from lecturer)

Project Phases & Deadlines

Phase	Focus	Deadline
Phase 1	Proposal & Wireframes	Saturday, 5th July 2026
Phase 2	Frontend Layout & Design (Interim)	Friday, 1st August 2026
Phase 3 + Final	PHP & MySQL Integration, Report & Viva	Sunday, 7th September 2026

Phase 1 – Proposal & Wireframes

Due: Saturday, 5th July 2026 | Submission: LMS Assignment Upload

Deliverables

1. Project Proposal Document

- Selected theme and project title
- Purpose and goals of the web application
- Target audience
- List of planned features (minimum three)
- Technology stack confirmation
- GitHub repository link (must be set up and shared)

2. Wireframes

- Wireframes for at least three pages (e.g. Home, Features, Contact)
- Use Figma, Balsamiq, or clearly legible hand-drawn scans
- Label all UI elements and page sections

Submission Format

- A single PDF containing the proposal and wireframe images
- Upload to LMS before the deadline

Phase 2 – Frontend Layout & Design

Due: Friday, 1st August 2026 | Submission: GitHub link pasted in LMS

Requirements

- Fully structured HTML pages (minimum three pages)
- CSS styling – consistent, clean, and themed
- Bootstrap 5 components (e.g. navbar, cards, carousel, modal)
- Responsive design – must adapt to mobile, tablet, and desktop
- Navigation bar and footer present on all pages

- At least three JavaScript features implemented (see list below)

Required JavaScript Features (minimum 3 of the following)

1. Dynamic Content Updates – toggling sections, filtering results, or live updates
2. Interactive Image Slider – manual or automatic image browsing
3. Form Validation – email validation, required fields, real-time feedback
4. Smooth Scrolling – scroll to sections via navigation links
5. Event Handling – hover effects, pop-ups, tooltips, or click events
6. Custom Animations – fade-ins, transitions, or other visual effects

Submission Format

- Push all work to your GitHub repository with meaningful commit messages
- Include a README.md describing the project theme and setup
- Paste your public GitHub repository link in the LMS submission field

Phase 3 – PHP & MySQL Integration (Final Submission)

Due: Sunday, 7th September 2026 | Submission: LMS (source code ZIP + report PDF + GitHub link)

1. Database Setup

- Use XAMPP or WAMP to create a local MySQL database
- Name the database based on your project theme (e.g. recipe_book, fitness_tracker)

2. Required Tables

User Table (users)

- Fields: id (AUTO_INCREMENT), username, email, password (hashed), created_at

Contact / Messages Table (messages)

- Fields: id, name, email, message, created_at

Theme-Specific Table(s)

- Example – Recipe App: recipes (id, title, ingredients, instructions, user_id)
- Example – Fitness Tracker: workouts (id, user_id, activity, duration, calories_burned)

3. User Authentication

- Registration page (register.php) – sign up with username, email, password
- Passwords must be hashed using password_hash() with PASSWORD_BCRYPT
- Login page (login.php) – validate credentials, start session, redirect to dashboard
- Call session_regenerate_id() after successful login
- Logout (logout.php) – destroy session cleanly

4. Contact Form (contact.php)

- A form where users can submit queries or messages
- Store submissions in the messages table
- Display a confirmation message upon successful submission

5. Backend–Frontend Integration

- All forms must submit and process data via PHP and MySQL
- JavaScript validation must remain intact alongside server-side validation
- Use prepared statements (PDO or MySQLi) to prevent SQL injection

6. Required Folder Structure

```
project/
├── css/
├── js/
├── images/
├── includes/
│   ├── db.php
│   └── functions.php
├── auth/
│   ├── register.php
│   ├── login.php
│   └── logout.php
├── contact.php
├── index.php
├── dashboard.php
├── README.md
└── database.sql
```

7. Final Submission Package

- Source code – ZIP file of the entire project folder
- database.sql – exported from phpMyAdmin
- Project Report – PDF (see Report section below)
- Updated GitHub repository link
- Upload all items to LMS before the deadline

Final Project Report

Submit the report as a PDF on LMS alongside the source code. It must cover the following sections:

1. Cover page – project title, theme, group members, index numbers, date
2. Project overview – purpose, target audience, and key features
3. System design – Database schema, folder structure
4. Implementation – description of each major feature with screenshots
5. Challenges and solutions – problems encountered and how they were resolved
6. Individual contribution – clear breakdown of who did what
7. Setup instructions – how to import the database and run using XAMPP
8. References

Viva

A viva will be conducted for all groups near the end of the semester. The date and format will be communicated in advance. Each group member will be questioned individually on their contribution and understanding of the project.

- Both members must be present at the scheduled viva
- Be prepared to explain any part of your code
- Be prepared to discuss design decisions, database structure, and PHP logic
- The viva allows you to demonstrate your knowledge beyond what is visible in the submitted work

Marking Scheme

The mini project carries 20% of the final assessment for ICT 1209. Marks are distributed as follows. Under each submission link you will find more details.

Component	Assessment Criteria	Weight
Phase 1 – Proposal & Wireframes	Clarity of proposal, completeness of wireframes, theme suitability	10%
Phase 2 – Frontend Layout & Design	HTML/CSS structure, Bootstrap usage, responsiveness, JavaScript features, code quality	20%
Phase 3 – PHP & Database Integration	Database design, authentication, contact form, backend logic, prepared statements, session handling	25%
Final Report	Coverage, clarity, ER diagram, screenshots, individual contribution section, setup instructions	15%
Viva	Individual understanding, ability to explain code and design decisions, response to questions	20%
Documentation & Code Quality	README, inline comments, Git commit history, folder structure compliance	10%
TOTAL		100%

The above percentages are out of the 20% CA component. Phase 1 and Phase 2 marks are awarded at submission. Phase 3, Report, and Viva marks are awarded after the final submission and viva session.

General Guidelines & Academic Integrity

- Work must be original. Do not copy code from the internet without understanding and proper attribution.
- Both group members are expected to contribute equally. Contribution will be verified at the viva.
- Use meaningful Git commit messages – commit history is part of the assessment.
- All submissions must be made via LMS before the deadlines stated. Late submissions may be penalized.
- Raise any questions during lectures or contact me before deadlines, not after.

Detailed phase-specific instructions will be shared on the LMS and in lectures. Ensure you are enrolled in the LMS course and check it regularly.